

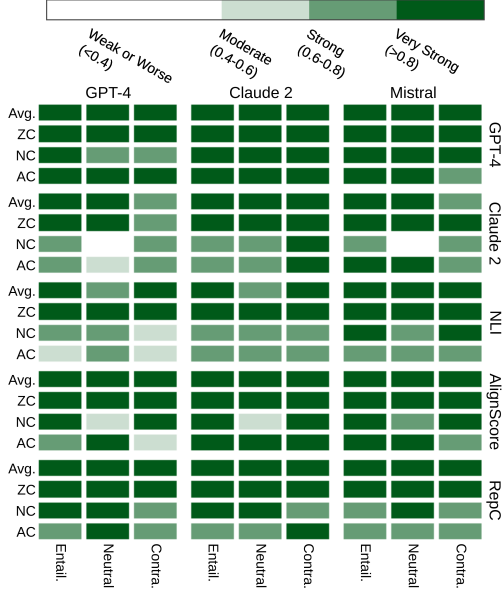


Figure 6: Spearman’s rank correlation coefficients between REFChecker and human evaluation. Results are grouped regarding extractors and checkers used. Results for each extractor (column) and checker (row) are arranged into a sub-matrix in the figure, with values for 4 context settings (one additional for average) and 3 ranking metrics.

averages. The results are reported in Figure 6.

We observe that the combination of Claude 2 + GPT-4 is the most competitive option with *very strong* correlations across all settings, benefiting from more powerful LLMs. Replacing the extractor with our Mistral alternative yields a marginal decline of performance, yet still maintains *very strong* correlations in most settings. The best non-proprietary combination is Mistral + NLI/AlignScore checker (356M), which achieve consistently *strong* correlations in all settings. The Mistral-RepC checker is robust against different extractors, owing to its stronger reasoning capability than small NLI-based checker. This result serves as a guide for choosing a configuration tailored to the user’s preferences. These preferences may include factors such as budget, deployment simplicity, specific settings, types of hallucination, privacy and requirements for open-source models.

6 Analysis and Discussion

We evaluate each component in REFChecker separately and discuss its limitations and corresponding future work in this part.

6.1 Evaluation on Extractors

To ensure precise hallucination detection, it requires precise claims that faithfully represent the

Extractor Model	GPT-4 Turbo Evaluation			Human Evaluation		
	Precision	Recall	F1	Precision	Recall	F1
GPT-4	97.2	92.5	94.2	98.2	92.2	94.8
Claude 2	95.5	91.8	93.0	97.4	94.5	95.7
Mistral	87.7	85.2	85.5	87.6	85.5	85.4

Table 3: Claim extraction evaluation by GPT-4 Turbo and human on 30 samples.

Extractor Model	Precision	Recall	F1	Speed (sec/iter)
Mistral	82.2	68.2	71.3	<u>1.7</u>
Mistral-SFT	<u>90.5</u>	84.8	86.4	1.7
Mixtral	86.7	80.2	81.6	5.7
Claude 2	89.8	<u>86.6</u>	<u>87.0</u>	6.9
GPT-4	92.4	88.6	89.3	8.7

Table 4: Automatic evaluation results of extractors. Mistral-SFT refers to our Mistral-based extractor after supervised fine-tuning. The other extractors directly prompt corresponding LLMs with two in-context examples. The best and the second best results are bolded and underlined, respectively.

facts in the original response. Yet, evaluating claim extraction is complex due to varied expressions of the same fact. To address this, we employ an automatic evaluation pipeline utilizing GPT-4 Turbo (gpt-4-1106-preview) to lessen the need for post-hoc human evaluation for each extractor.

We employed GPT-4 Turbo to label each extracted claim as True/False, indicating faithfulness to the original semantics. Additionally, we tasked it with completing missing claims, enabling automatic calculation of **precision**, **recall**, and **F1** score. To validate results, we conducted a human evaluation on 30 random samples, ensuring agreement between human annotators and the model. The comparison in Table 3 demonstrates strong alignment between human and automatic evaluations, achieving 93.7% agreement on precision and 91.9% on recall.

Leveraging the reliability of our automatic evaluation pipeline, we evaluated the performance of four extractors (see Table 4). Our open-source Mistral extractor achieves performance comparable to Claude 2 extractor with faster inference speed and no need for API tokens.

6.2 Evaluation on Checkers

As described in Sec. 4.2, the baseline checkers we include in the evaluation are RoBERTa-NLI, AlignScore, GPT-4 and Claude 2. The Mistral-based checkers we include are zero-shot prompted, one-shot prompted, LoRA fine-tuned and RepC-LE variants. The training and development data

Models	Average of three settings		Zero-context (NQ)		Noisy-context (MS MARCO)		Accurate-context (databricks-dolly-15k)	
	Accuracy	Macro-F1	Accuracy	Macro-F1	Accuracy	Macro-F1	Accuracy	Macro-F1
Baseline Checkers								
RoBERTa-NLI	76.56	55.88	74.06	<u>69.90</u>	78.36	46.67	77.27	51.06
AlignScore	78.85	<u>59.45</u>	73.40	70.28	78.86	<u>50.42</u>	84.30	<u>57.66</u>
GPT-4	74.77	59.80	67.46	66.10	76.67	55.49	80.17	57.80
Claude 2	51.98	36.55	43.42	42.90	40.35	25.89	72.18	40.87
Mistral-based Checkers								
zero-shot	69.43	46.64	70.83	61.10	71.75	43.01	65.72	35.81
1-shot	76.68	50.66	65.44	63.25	81.23	42.18	83.38	46.56
LoRA-sft-n4000	77.84	57.98	77.43	73.64	79.21	<u>50.29</u>	76.89	50.00
RepC-LE-svm-n1000-e1000	79.03	<u>60.05</u>	77.98	<u>73.53</u>	79.56	51.29	79.54	<u>55.34</u>
RepC-LE-nn-n2000-e2000	81.27	60.80	75.23	71.98	82.08	47.56	86.50	62.86

Table 5: Checker evaluation results on 11k human annotated claim triplets. In Mistral-based checkers, the model names start with the variant types, eg. LoRA-sft indicates the LoRA fine-tuned variant and RepC-LE-nn indicates the representation based classification variant using layer ensemble with 2-layer MLP as the classifier. Here “nxxx” and “exxx” indicates the number of training samples and ensemble learning samples. Due to the space limitation, we do not include all variant results here, please refer to Table 8 of Appendix B.2 for full results.

of these variants are 4k samples from the ANLI dataset (Nie et al., 2020). We evaluate their performance using the 11k manually annotated claim triplets. The evaluation metric is accuracy and macro-f1 score over 3-way classification.

Table 5 shows the evaluation results. Among the baseline checkers, AlignScore is a strong competitor to GPT-4, and Claude 2 has a significant gap. We found Claude 2’s neutral F1 score is very low (less than 20%) (cf. Table 9 in Appendix B.2), with a tendency to flag neutral claims as contradiction, as a result of biasing towards its own internal knowledge when asked to perform checking (cf. Appendix D).

Besides, the Mistral-based checkers can often gives the best performance, though there does not yet exist a single winner across the board. The weakness of Mistral-based checkers lies in the NC setting. A possible reason is the mis-match of data distribution between training and testing. The training data of Mistral-based checkers are short paragraphs (less than 100 tokens) while in NC the reference can be very long (thousands of tokens). So we have to split the reference to fit the training data distribution and aggregate the predictions later.

However, there are clear gaps between the performance of the checkers on NC and AC in contrast to their performance on ZC, which suggests ample room of improvement for the checkers.

6.3 Future Work

There remain several limitations of REFCHECKER that require addressing: **a)** The triplet format of claims, while effectively breaking down LLM responses into finer granularity, can be overly restric-

tive and may not have the flexibility to cover important semantics (consider (Trump, president of, US), which is factual in 2018, but non-factual in 2022). **b)** While extraction is relatively simple with the outstanding comprehension capability of LLMs, there exists a large improvement space for more powerful checkers. **c)** REFCHECKER has a rudimentary support for source attribution (cf. Appendix B.3). Better source attribution is critical to lend explainable and provide training signal to mitigate hallucination. **d)** We found that model-based checkers may exhibit bias towards internal knowledge, declaring a neutral claim to be entailment or contradiction (cf. Table 9 and Appendix D). This requires we inject some forms of “source control” in LLMs. **e)** In actual deployment cases, we found users ask for stronger customizability (e.g. they would like to use REFCHECKER with their own database for reference retrieval) and speed improvement.

7 Conclusion

We introduce REFCHECKER, a unified framework for detecting hallucination in LLM responses. REFCHECKER operates at the level of knowledge triplets, termed claim-triplets, extracted from LLM responses, allowing for fine-grained detection. These claim-triplets are then evaluated against references to determine their hallucination categories. An automated pipeline pairs an extractor and a checker to identify potential hallucinations, calibrated to match human annotations, yielding superior performance compared to prior methods.

Limitations

a) We have anecdote evidence that sometimes hallucination is due to reasoning and limited context-window. These advanced form of hallucination is difficult to be dealt with using triplet which bias towards local contexts. **b)** At its current form, REFCHECKER primarily focuses on plain text in general domains. Exploring extensions to include various data formats (table, code, math, etc.) and specific domains (business, medical, legal, etc.) is worthy of consideration.

Ethics Statement

We contend that REFCHECKER poses no negative ethical implications for the public; rather, it holds the potential for positive impact by enabling the identification of non-factual content within the responses generated by large language models (LLMs). This capability contributes to the cultivation of responsible AI practices for the benefit of society.

In this study, we utilized a variety of scientific resources to conduct our research and aim to contribute additional artifacts to the community. Specifically, to curate the benchmark dataset, we sample 100 examples from each of the following datasets:

- The development set of NaturalQuestions dataset, which is under Creative Commons Share-Alike 3.0 License.
- The validation set of MS MARCO dataset, which is under Creative Commons Attribution 4.0 International License.
- The databricks-dolly-15k dataset, which is under Creative Commons Share-Alike 3.0 License.

These datasets are publicly accessible and utilize English language corpora. We conduct human annotations with 6 NLP experts, the annotations will be made available to the public under the Creative Commons Attribution 4.0 International License.

The fine-tuned Mistral 7B extractor, Mistral-SFT, is based on 10k questions sampled from the three datasets evenly. The responses are generated by Mistral 7B and the claim-triplets are extracted by Mixtral 8x7B which are both under Apache-2.0 License. The RepC checker is also based on Mistral 7B and is trained with the ANLI dataset which

is under Creative Commons-Non Commercial 4.0 License. The fine-tuned models will be released to the public under Apache-2.0 License.

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